

AKSHA, Russian Federation

WMO: 309570

Lat: 50.267N

Lon: 113.267E

Elev: 2402

StdP: 13.46

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.1	-27.8	-36.4	0.8	-30.9	-32.1	1.0	-26.6	20.7	7.0	17.9	5.8	2.8	190	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	87.8	63.2	83.8	62.3	80.0	61.2	67.3	79.4	65.6	76.9	64.0	74.8	8.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	97.0	70.9	62.0	90.9	69.2	60.3	85.5	67.7	33.3	79.9	31.9	76.9	30.6	74.7	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.3	18.4	16.0	DB	-39.9	94.6	7.5	5.2	-45.3	98.3	-49.7	101.3	-53.9	104.3	-59.4	108.0	(4)
(5)			WB	-38.6	71.1	6.2	2.1	-43.1	72.6	-46.7	73.8	-50.2	75.0	-54.8	76.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.8	-7.0	1.4	18.5	36.0	49.0	61.8	66.3	62.2	49.5	32.8	12.2	-3.2		
	DBStd	27.04	9.39	10.68	11.41	8.20	7.25	6.54	4.95	5.96	7.18	8.64	11.24	10.36		
	HDD50	8055	1767	1360	976	425	108	3	0	2	95	535	1133	1651		
	HDD65	12282	2232	1780	1441	870	496	134	40	125	466	999	1583	2116		
	CDD50	1406	0	0	0	5	78	357	505	379	80	2	0	0		
	CDD65	157	0	0	0	0	1	38	80	37	1	0	0	0		
	CDH74	2599	0	0	0	7	129	815	1053	542	53	0	0	0		
	CDH80	849	0	0	0	1	33	267	372	168	8	0	0	0		
	WSAvg	5.9	5.1	5.3	6.3	7.6	8.0	6.1	5.2	5.0	5.6	5.8	5.4	4.9		
	PrecAvg	15.2	0.1	0.1	0.2	0.5	1.3	2.0	4.4	3.9	1.7	0.4	0.2	0.2		
	PrecMax	24.9	0.3	0.4	0.6	1.6	3.4	4.7	10.4	11.7	5.4	1.3	0.7	0.6		
(15) Precipitation	PrecMin	9.3	0.0	0.0	0.0	0.0	0.1	0.1	1.2	0.9	0.1	0.0	0.0	0.0		
	PrecStd	4.2	0.1	0.1	0.2	0.4	0.9	1.1	1.9	2.2	1.2	0.3	0.2	0.1		
	DB	23.8	37.3	55.4	71.7	85.0	92.1	95.3	90.0	80.4	66.3	44.9	27.1			
	MCWB	19.5	28.6	40.4	49.3	57.3	63.5	65.9	64.9	58.6	47.8	35.1	22.6			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	17.6	30.3	48.4	64.3	77.9	86.7	88.6	84.5	73.9	58.5	38.3	21.6			
	2% MCWB	14.6	24.3	36.1	45.6	54.4	61.7	64.6	63.6	55.6	43.8	31.0	18.0			
	5% DB	12.7	25.1	42.8	58.7	72.4	82.6	83.6	79.9	69.1	52.4	33.6	16.9			
	5% MCWB	10.3	20.4	32.7	42.6	51.7	60.7	63.9	62.8	53.3	40.7	28.0	14.3			
	10% DB	8.7	20.3	37.4	53.4	67.4	78.5	79.4	75.5	64.5	47.6	29.6	12.8			
	10% MCWB	6.9	16.6	29.3	39.6	48.9	59.2	63.1	61.6	51.2	37.8	25.0	10.7			
	0.4% WB	19.6	29.1	41.3	50.4	58.5	67.4	70.0	69.1	61.1	48.5	35.0	23.0			
	0.4% MCDB	23.7	36.9	55.5	69.9	80.6	81.6	82.3	82.9	75.0	65.5	44.0	26.8			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	14.6	24.3	36.5	46.2	55.4	64.8	68.1	66.3	57.9	44.2	31.2	18.2			
	2% MCDB	17.7	30.0	47.7	63.3	75.8	78.5	80.6	77.1	68.4	56.7	37.7	21.5			
	5% WB	10.4	20.5	32.9	43.1	52.8	62.9	66.5	64.6	55.4	41.2	28.1	14.4			
	5% MCDB	12.6	25.1	42.4	57.8	69.7	76.1	78.1	75.3	65.2	51.1	33.4	16.8			
	10% WB	6.8	16.7	29.6	40.1	50.2	61.2	64.9	62.9	53.0	38.3	25.0	10.7			
	10% MCDB	8.6	20.3	37.2	52.5	65.6	74.5	75.5	72.9	62.6	46.7	29.4	12.7			
	MDBR	23.4	29.2	28.2	25.7	28.0	26.7	23.2	23.4	25.1	23.2	21.8	21.1			
	5% DB	25.5	32.9	33.3	36.5	40.1	36.6	33.0	32.1	34.2	32.5	24.4	22.5			
(36) Mean Daily Temperature Range	MCWBR	22.7	27.3	22.0	20.8	19.6	14.9	12.8	13.7	16.9	19.1	17.8	19.5			
	MCDBR	25.2	32.7	32.3	34.4	35.5	28.4	24.7	25.4	28.4	30.6	24.0	22.4			
	MCWBR	22.6	27.1	21.5	20.1	17.9	13.0	11.4	12.2	14.3	18.4	17.6	19.5			
	MCDBR	22.6	27.1	21.5	20.1	17.9	13.0	11.4	12.2	14.3	18.4	17.6	19.5			
(40) Clear-Sky Solar Irradiance	taub	0.249	0.274	0.313	0.380	0.457	0.447	0.419	0.381	0.334	0.299	0.270	0.254			
	taud	2.297	2.271	2.263	2.171	1.978	2.130	2.273	2.362	2.435	2.451	2.367	2.256			
	Ebn at Noon	255	276	282	272	254	259	265	270	273	265	244	231			
	Edh at Noon	21	28	34	43	54	47	40	35	29	23	20	19			
(44) All-Sky Solar Radiation	RadAvg	516	880	1375	1710	1890	1940	1786	1589	1283	858	524	397			
	RadStd	15	27	44	123	109	129	136	121	92	43	27	24			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page